AnyAttack: Towards Large-scale Self-supervised Adversarial Attacks on Vision-language Models

Anonymous Author(s)

Abstract

Targeted adversarial attacks on vision-language models are typically constructed with per-example optimization, explicit target labels, or textual supervision, which limits scalability and weakens transfer to black-box systems. We present AnyAttack, a transfer-based attack that reframes targeted multimodal evasion as an amortized generation problem. AnyAttackfirst pre-trains a perturbation decoder in a self-supervised way on LAION-400M by learning to map frozen CLIP image embeddings to bounded perturbations that make randomly chosen carrier images align with the target image embedding, without using labels or text prompts. The same generator is then fine-tuned for downstream multimodal tasks with lightweight task-specific objectives. This pre-train-then-adapt design yields a single-forward-pass attack at inference time. On the provided evaluation suite, AnyAttacksubstantially improves targeted transfer on multiple vision-language settings: on MSCOCO retrieval it reaches 35.88 R@Mean on CLIP ViT-B/16, 33.66 on ViT-L/14, and 33.18 on ViT-L/14×336, outperforming the strongest baseline in each case by 15.02 to 18.54 points; on SNLI-VE it improves attack accuracy to 44.8% versus 24.8% for the best baseline; on MSCOCO captioning it increases CIDEr to 20.3 on InstructBLIP, 13.3 on BLIP2, 13.0 on BLIP, and 5.3 on MiniGPT-4; and it also transfers to commercial systems, reaching 31% attack success on Gemini and 38% on GPT. These results suggest that large-scale self-supervised pre-training can make targeted multimodal attacks both more scalable and more transferable.

1. Introduction

Vision-language models (VLMs) such as CLIP, BLIP, BLIP-2, InstructBLIP, and MiniGPT-4 have become common building blocks for retrieval, captioning, classification, and instruction-following systems [2, 8, 9, 11, 23]. Their deployment surface is correspondingly broad, which makes adversarial robustness a practical security concern rather than a purely academic one [5, 10, 20]. In particular, targeted attacks are especially concerning: instead of merely degrading performance, they can steer a system toward attacker-chosen semantics.

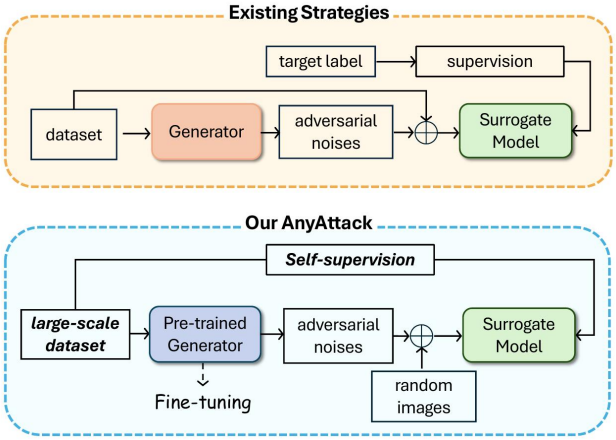

Figure 1. Comparison between prior targeted attack paradigms and the proposed self-supervised formulation of AnyAttack. Prior approaches depend on explicit target supervision, whereas AnyAttacklearns a target-conditioned perturbation generator that can be applied to arbitrary carrier images.

Existing targeted attacks on VLMs are still difficult to scale. Many methods rely on iterative optimization against a surrogate model, require explicit target labels or target text, or optimize a universal perturbation for a narrow setting [18, 20–22]. These choices make attacks expensive to generate, tightly coupled to specific tasks, and less suitable for black-box transfer across heterogeneous downstream models. Similar limitations have also been observed in targeted transfer attacks for image classification, where success often depends on aggressive per-example optimization and carefully engineered transformation heuristics [15–17, 19].

This paper explores a different viewpoint: targeted multimodal attacks can be amortized. Instead of solving a new optimization problem for every target, we pre-train a perturbation generator once at large scale, then adapt it with lightweight downstream objectives. The central idea is to learn a foundation model for adversarial noise: given a target image embedding, the generator predicts a bounded perturbation that, when added to a random carrier image,

Strategy	Formulation
Target label supervision	$\min \mathcal{L}(f_s(x+\delta), y_t), \quad \text{s.t. } y_t \neq y$
Target image supervision	$\min \mathcal{L}(f_s(x+\delta), f_s(x_t)), \quad \text{s.t. } x_t \neq x$
AnyAttack (ours)	$\min \mathcal{L}(f_s(\delta + x_r), f_s(x))$

Table 1. The formulation of different attack strategies. The existing strategies rely on explicit target supervision, whereas our AnyAttack is unsupervised.

makes the attacked image semantically align with the target. Because this alignment objective is defined entirely in the representation space of a frozen encoder, pre-training can be performed without labels or text supervision.

The resulting method, AnyAttack, has two stages. In pre-training, a decoder conditioned on frozen CLIP image features is trained with a self-supervised contrastive objective on LAION-400M [13]. In fine-tuning, the same decoder is adapted to downstream tasks using either a cosine-alignment loss or a bidirectional contrastive loss, optionally augmented with auxiliary frozen encoders to improve cross-model transfer. At inference time, generating an attack requires only a single forward pass through the encoder-decoder pair.

Our experiments, grounded in the supplied quantitative artifacts, show that this formulation is effective across multiple tasks. AnyAttackimproves targeted transfer on MSCOCO retrieval, SNLI-VE classification, and MSCOCO captioning, and also transfers to commercial VLM APIs. The ablation results further indicate that large-scale pre-training is critical, auxiliary encoders materially improve transfer, and the bidirectional objective is preferable for retrieval. Because attack generation is amortized, AnyAttackalso enjoys favorable time and memory cost relative to iterative baselines.

Our contributions are threefold. First, we introduce a self-supervised pre-training objective for targeted adversarial perturbation generation that removes the need for labels or text prompts during large-scale training. Second, we show that a pre-train-then-fine-tune perturbation generator can transfer across multiple VLM architectures and downstream tasks. Third, we provide empirical evidence that such a generator exposes meaningful vulnerabilities not only in open-source VLMs but also in commercial systems.

2. Related Work

Adversarial transfer attacks. Transfer-based attacks exploit the cross-model portability of adversarial perturbations [5, 10]. In image classification, a large literature improves transferability through momentum, input diversity, local transformations, and multi-image mixing [14, 15, 17]. Targeted transfer remains substantially harder than non-targeted transfer, which has motivated methods such as Self-Universality and Beyond-ImageNet Attack to stabilize perturbations and reduce surrogate overfitting [16, 19]. AnyAttackshares the goal of strong black-box transfer, but replaces iterative attack-time optimization with an amortized generator learned at scale.

Adversarial attacks on vision-language models. Multimodal attacks have mainly focused on attacking a fixed downstream task or jointly perturbing images and text [18]. Recent work has examined downstream-agnostic or universal perturbations for contrastive encoders and pretrained VLMs [21, 22], as well as black-box robustness of large VLMs [20]. These studies establish that VLMs are vulnerable, but they typically optimize task-specific perturbations, sample-agnostic universal patches, or surrogate-specific iterative attacks. AnyAttackinstead targets arbitrary image-specified semantics with a single target-conditioned generator, which is trained without supervision during the large-scale stage and then adapted to downstream tasks.

Self-supervised and large-scale representation learning. Contrastive learning has shown that rich semantic representations can be learned from instance discrimination objectives without labels [1, 6]. In multimodal learning, CLIP and related models demonstrated that large-scale web supervision produces highly transferable image-text representations [7, 11]. Our method borrows the scalability of this paradigm, but uses it for attack generation rather than recognition: the contrastive objective is applied to attacked carrier images and their target embeddings so that the perturbation generator learns how to inject transferable semantics into arbitrary carriers.

3. Method

Problem setting. We consider a black-box transfer setting. The attacker does not access victim parameters and does not optimize against the victim at test time. Instead, the attacker trains a perturbation generator using public surrogate encoders and then transfers the generated adversarial images to downstream VLMs. The attack is targeted in the sense that, given a target image x_t, the adversarial image should induce victim outputs aligned with the target semantics rather than the carrier content.

Attack parameterization. Let E denote a frozen CLIP image encoder [11]. For a target image x_t, we compute its embedding $z_t = E(x_t) \in \mathbb{R}^{512}$. A decoder F maps z_t to a perturbation $\delta = F(z_t)$, followed by element-wise clipping to the ℓ_∞ budget $\epsilon = 16/255$. The perturbation is added to a carrier image x_r:

$$\hat{x} = \text{clip}(x_r + \text{clip}(F(E(x_t)), -\epsilon, \epsilon), 0, 1). \quad (1)$$

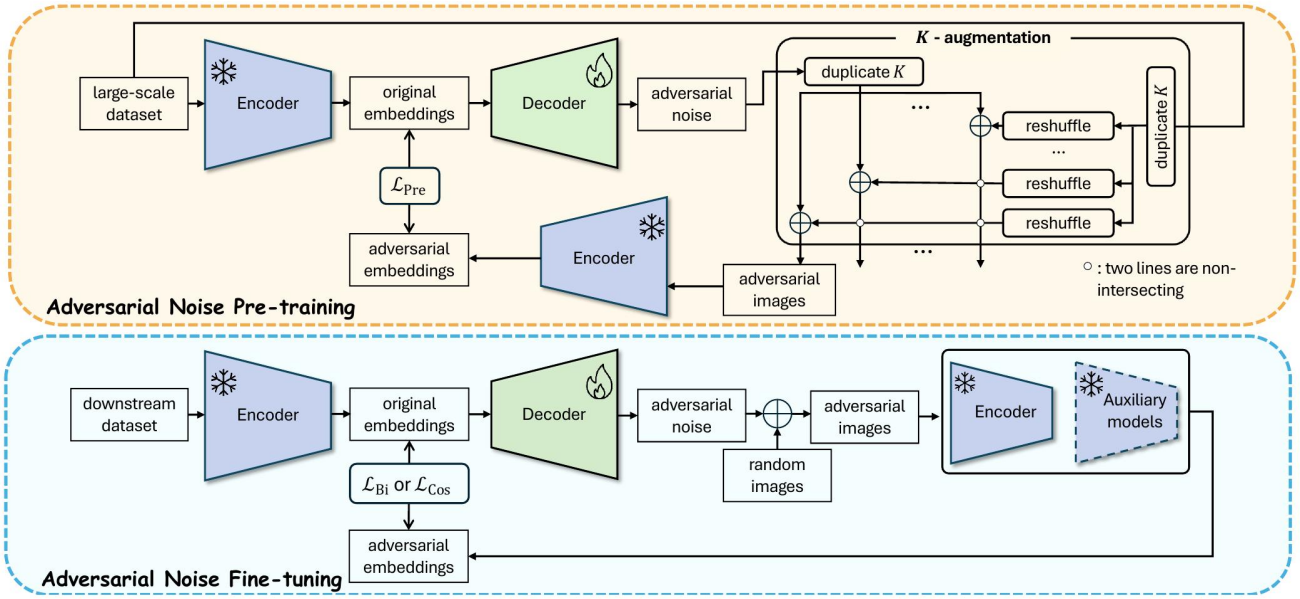

Figure 2. Overview of AnyAttack. A perturbation decoder is first pre-trained in a self-supervised manner using frozen CLIP image embeddings and shuffled carrier images, then fine-tuned on downstream multimodal tasks with task-appropriate objectives. At test time, attacks are generated in one forward pass.

The same generator is used during pre-training, fine-tuning, and inference. Unlike a universal perturbation, δ is target-conditioned; unlike iterative attacks, it is generated in one pass.

Generator architecture. The decoder takes a 512-dimensional CLIP embedding and projects it to a $256 \times 14 \times 14$ feature map with a linear layer. It then alternates residual blocks and upsampling blocks to progressively increase spatial resolution from 14 to 28, 56, 112, and 224. Each residual block uses 3×3 convolutions, batch normalization, LeakyReLU activations, and EfficientAttention modules, and the final layer maps 16 channels to RGB. The perturbation bound is enforced outside the decoder by clipping. CLIP normalization is applied inside the frozen encoder.

Self-supervised pre-training. The large-scale stage is trained on LAION-400M [13] with frozen CLIP ViT-B/32 features. For each mini-batch, images are duplicated $K{=}5$ times and shuffled to create carrier-target mismatches across duplicates. This produces attacked carriers whose semantics must move toward the original target embeddings rather than reconstruct the carrier. Let $\hat{z}_i = E(\hat{x}_i)$ be the embedding of the attacked carrier corresponding to target x_i. We optimize an InfoNCE-style objective [1]:

$$\mathcal{L}_{\text{pre}} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(\text{sim}(\hat{z}_i, z_i)/\tau)}{\sum_{j=1}^{B} \exp(\text{sim}(\hat{z}_i, z_j)/\tau)}, \quad (2)$$

where $\text{sim}(\cdot,\cdot)$ is cosine similarity and τ is annealed exponentially from 1.0 to 0.07 over the first 10k steps. Pre-training runs for 520k steps with AdamW, weight decay 0.01, learning rate 10^{-4}, and cosine warm restarts.

This objective is self-supervised: it never uses category labels or text prompts. The only supervision is that attacked carriers should align with their source image embeddings more than with other targets in the batch. Intuitively, the generator learns to transplant target semantics into arbitrary carriers while respecting a small perturbation budget.

Task-specific fine-tuning. The pre-trained decoder is then adapted for downstream tasks using target images from the downstream benchmark and random carrier images from the ImageNet-1K validation set [12]. Fine-tuning lasts 20 epochs with batch size 800. We consider two objectives.

For general multimodal classification and captioning settings, we use cosine alignment:

$$\mathcal{L}_{\text{cos}} = 1 - \text{sim}(E_m(\hat{x}), E_m(x_t)), \quad (3)$$

where E_m denotes a frozen surrogate encoder.

For retrieval, we use a bidirectional contrastive objective $\mathcal{L}_{\text{bi}}$ with fixed temperature 0.07, implemented over attacked-target positive pairs and in-batch negatives so that the attacked image is simultaneously optimized for both retrieval directions. In practice, the total fine-tuning loss sums

the chosen task loss across the main encoder and two auxiliary frozen encoders with equal weight:

$$\mathcal{L}_{\text{ft}} = \sum_{m \in \mathcal{M}} \mathcal{L}_m, \tag{4}$$

where $\mathcal{M}$ contains CLIP-based surrogates and, when enabled, a torchvision ViT-B/16 [3] and EVA-02 Large Patch14 [4]. We refer to these variants as "w/ Aux" in the experiments.

Why this works. AnyAttackis designed around two hypotheses. First, a perturbation generator trained across massive carrier-target mismatches can learn transferable semantic attack primitives instead of overfitting to one surrogate or one task. Second, fine-tuning with multiple frozen encoders regularizes the perturbation toward features shared across models, which should improve black-box transfer. The experimental section supports both hypotheses: pre-training and auxiliary encoders each contribute substantial gains.

4. Experiments

Benchmarks and protocol. We evaluate the supplied experimental results on three settings: image-text retrieval on MSCOCO, multimodal classification on SNLI-VE, and image captioning on MSCOCO. The provided setup uses the top 1,000 downstream images for each task and the top 1,000 ImageNet validation images as carriers. Victim models include CLIP ViT-B/16, ViT-L/14, and ViT-L/14×336 for retrieval; BLIP, BLIP2, InstructBLIP, and MiniGPT-4 for captioning; and a multimodal classifier on SNLI-VE. We also report transfer to two commercial systems on a 100-image evaluation set.

Baselines. We compare against AttackVLM [20] and transfer-based attacks adapted from prior classification literature, including Self-Universality [16] and structure-preserving transformation attacks [15]. The supplied tables report cosine- and MSE-based adaptations of these baselines. Higher values indicate stronger targeted transfer under the task-specific metric: recall-based retrieval, classification accuracy, captioning metrics computed against target outputs, and API-level attack success rate on commercial systems.

Implementation details. Unless otherwise noted, the perturbation budget is $\epsilon = 16/255$. Pre-training uses CLIP ViT-B/32 embeddings from 224×224 images, while auxiliary fine-tuning uses ViT-B/16 at 224×224 and EVA-02 Large Patch14 at 448×448. We report the two downstream fine-tuning variants used in the artifacts: AnyAttack-Cos and AnyAttack-Bi, each with or without auxiliary encoders.

4.1. Main Results

Image-text retrieval. Table 2 shows that AnyAttackis consistently the strongest attack on MSCOCO retrieval. The best variant, AnyAttack-Bi w/ Aux, reaches 35.88 R@Mean on CLIP ViT-B/16, 33.66 on ViT-L/14, and 33.18 on ViT-L/14×336. The strongest non-AnyAttackbaseline reaches 20.86, 15.22, and 14.64 respectively, so the absolute gains are 15.02, 18.44, and 18.54 points. These improvements hold across both text retrieval and image retrieval metrics, indicating that the perturbations transfer to multiple CLIP scales rather than overfitting to one victim.

Multimodal classification. On SNLI-VE, Table 3 shows a large gap between the best baseline and the proposed method. AnyAttack-Cos w/ Aux reaches 44.8% accuracy, compared with 24.8% for the strongest baseline and 17.5% for AnyAttack-Cos without auxiliaries. This result is notable because the task is semantically farther from the contrastive pre-training setup than retrieval, yet the pre-trained generator still transfers effectively after lightweight adaptation.

Image captioning. Table 4 reports targeted captioning transfer on four open-source VLMs. The auxiliary variant improves CIDEr to 20.3 on InstructBLIP, 13.3 on BLIP2, 13.0 on BLIP, and 5.3 on MiniGPT-4. Compared with the strongest baseline for each model, the gains are +8.5, +3.8, +0.6, and +2.5 CIDEr respectively. AnyAttackis therefore effective not only on contrastive retrieval models, but also on generative VLMs with different architectures and training recipes.

4.2. Transfer to Commercial Systems

To test whether the learned perturbations survive deployment mismatches, we evaluate on commercial VLMs using the prompt "Evaluate the relationship between the given image and text" and count responses rated as "highly relevant" or "partially relevant" as successful attacks. Table 5 shows that AnyAttack-Cos w/ Aux reaches 31% attack success rate on Google Gemini and 38% on OpenAI GPT, outperforming the other baselines in the provided evaluation. Figure 3 gives qualitative examples, illustrating that the perturbations can transfer even when the victim interface and internal model are both unknown.

5. Analysis

Ablation on pre-training, objectives, and auxiliaries. The ablation plot in Figure 4 supports three conclusions. First, pre-training is critical: the pre-trained generator fine-tunes substantially better than training from scratch, suggesting that the large-scale self-supervised stage learns reusable attack priors. Second, the auxiliary encoders

Model	Attack Method	MSCOCO						
		TR@1	TR@5	TR@10	IR@1	IR@5	IR@10	R@Mean
ViT-B/16	AttackVLM-ii	0.4	1.0	1.4	0.24	1.08	2.16	1.05
	AttackVLM-it	0.2	1.4	1.8	0.16	1.16	2.12	1.14
	SASD-WS-Cos	6.0	17.0	24.8	9.08	24.39	34.55	19.30
	SASD-WS-MSE	4.8	18.4	25.6	8.20	25.87	35.15	19.67
	SU-Cos	6.8	20.4	27.8	11.11	25.70	33.34	20.86
	SU-MSE	6.8	20.6	27.0	10.83	25.10	32.62	20.49
	AnyAttack-Cos	8.6	21.2	29.6	10.80	27.59	37.50	22.55
	AnyAttack-Bi	12.2	26.2	33.8	12.63	31.71	40.86	26.23
	AnyAttack-Cos w/ Aux	8.4	24.8	33.0	11.59	32.10	44.98	25.81
	AnyAttack-Bi w/ Aux	**14.8**	**36.8**	**48.0**	**17.59**	**42.02**	**56.05**	**35.88**
ViT-L/14	AttackVLM-ii	0.2	1.0	1.6	0.24	0.60	1.32	0.83
	AttackVLM-it	0.4	0.8	1.4	0.12	0.76	1.48	0.83
	SASD-WS-Cos	3.8	11.6	18.8	7.20	18.43	26.35	14.36
	SASD-WS-MSE	5.4	14.6	20.6	6.00	18.23	26.47	15.22
	SU-Cos	3.0	10.4	13.2	6.19	14.99	20.07	11.31
	SU-MSE	3.4	11.2	17.4	6.63	15.27	19.79	12.28
	AnyAttack-Cos	3.8	14.0	22.8	7.36	20.71	27.55	16.04
	AnyAttack-Bi	4.8	16.0	23.6	8.20	22.31	29.11	17.34
	AnyAttack-Cos w/ Aux	9.4	24.6	37.0	11.51	32.62	48.18	27.22
	AnyAttack-Bi w/ Aux	**12.0**	**34.0**	**47.4**	**15.67**	**39.34**	**53.54**	**33.66**
ViT-L/14 $\times$ 336	AttackVLM-ii	0.2	0.6	1.6	0.16	1.12	2.04	0.95
	AttackVLM-it	0.2	0.6	1.8	0.32	0.96	1.76	0.94
	SASD-WS-Cos	2.8	10.8	16.4	6.52	18.31	26.19	13.50
	SASD-WS-MSE	4.4	13.6	19.2	6.72	18.23	25.71	14.64
	SU-Cos	2.4	8.0	11.2	4.88	13.79	18.39	9.78
	SU-MSE	3.6	8.2	13.2	6.40	14.51	19.19	10.85
	AnyAttack-Cos	4.6	11.0	16.6	5.96	17.67	24.23	13.34
	AnyAttack-Bi	3.6	14.4	19.0	7.64	19.79	26.83	15.21
	AnyAttack-Cos w/ Aux	9.0	23.2	37.2	11.68	34.03	47.62	27.12
	AnyAttack-Bi w/ Aux	**12.0**	**33.2**	**46.8**	**14.79**	**39.22**	**53.06**	**33.18**

Table 2. The retrieval performances on the MSCOCO dataset under different attacks. TR@1, TR@5, and TR@10 measures text retrieval performance, while IR@1, IR@5, and IR@10 measures image retrieval performance. R@Mean is the average of all retrieval metrics. Our proposed methods are *italicized*, the best results are highlighted in **bold**, and the second-best results are underlined.

Attack Method	**Accuracy**
AttackVLM-ii	6.5
AttackVLM-it	6.3
SASD-WS-Cos	24.3
SASD-WS-MSE	24.8
SU-Cos	13.7
SU-MSE	13.6
AnyAttack-Cos	17.5
AnyAttack-Cos w/ Aux	**44.8**

Table 3. Attack performance comparison on the SNLI-VE dataset for multimodal classification.

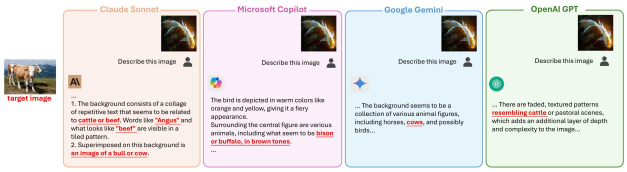

Figure 3. Examples of targeted transfer on commercial VLMs. The supplied examples show that attacked carriers can be judged as semantically relevant to the target content even in black-box API settings.

yield clear gains, consistent with the hypothesis that multi-encoder supervision improves transferability. Third, the bidirectional objective is superior to cosine alignment for retrieval, which matches the symmetric structure of the retrieval task.

Efficiency. Figure 5 compares runtime and memory usage. Although the exact numeric values are shown in the figure rather than the tables, the trend is clear: AnyAttackis substantially cheaper than iterative baselines because attack

Model	Attack Method	SPICE	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
InstructBLIP	AttackVLM-ii	1.4	38.9	5.4	8.7	28.6	3.4
	AttackVLM-it	1.3	39.1	5.4	8.7	28.8	3.3
	SASD-WS-Cos	3.4	43.9	7.2	10.5	32.2	10.9
	SASD-WS-MSE	3.2	44.6	7.0	10.8	32.4	11.8
	SU-Cos	1.9	40.7	6.0	9.3	29.9	5.3
	SU-MSE	1.9	40.9	6.5	9.5	29.9	6.0
	AnyAttack-Cos	2.3	41.5	5.9	9.5	30.2	7.0
	AnyAttack-Cos w/ Aux	**4.7**	**46.5**	**7.5**	**12.2**	**33.6**	**20.3**
BLIP2	AttackVLM-ii	1.2	39.6	5.3	8.7	29.0	3.6
	AttackVLM-it	1.2	39.6	5.4	8.7	29.3	3.5
	SASD-WS-Cos	2.6	43.0	6.3	10.2	32.0	9.3
	SASD-WS-MSE	2.8	42.8	6.5	10.2	31.7	9.5
	SU-Cos	1.6	40.9	5.6	9.2	30.1	4.7
	SU-MSE	1.6	40.8	5.9	9.2	30.1	5.0
	AnyAttack-Cos	1.8	41.3	5.2	9.6	30.9	5.6
	AnyAttack-Cos w/ Aux	**3.3**	**44.2**	**6.0**	**11.0**	**32.4**	**13.3**
BLIP	AttackVLM-ii	1.3	39.8	5.0	8.8	29.9	3.4
	AttackVLM-it	1.2	39.7	4.8	8.7	29.7	3.2
	SASD-WS-Cos	3.3	43.8	6.9	10.7	32.3	11.9
	SASD-WS-MSE	**3.4**	43.8	6.9	10.8	32.3	12.4
	SU-Cos	2.6	43.0	6.5	10.1	31.8	8.7
	SU-MSE	2.6	42.4	6.4	9.9	31.6	8.4
	AnyAttack-Cos	2.2	41.6	6.0	9.5	31.1	6.1
	AnyAttack-Cos w/ Aux	**3.4**	**44.4**	**7.1**	**11.1**	**32.8**	**13.0**
Mini-GPT4	AttackVLM-ii	1.6	29.5	2.3	9.3	24.3	1.6
	AttackVLM-it	1.5	29.2	2.3	9.4	24.5	1.5
	SASD-WS-Cos	2.8	30.5	2.4	10.9	25.2	2.6
	SASD-WS-MSE	3.1	30.5	2.9	10.9	25.7	2.8
	SU-Cos	2.0	29.5	2.9	9.9	24.8	1.9
	SU-MSE	2.2	30.3	2.9	9.9	25.1	2.2
	AnyAttack-Cos	2.5	30.5	2.4	10.3	25.2	1.9
	AnyAttack-Cos w/ Aux	**4.6**	**32.5**	**4.0**	**12.4**	**27.3**	**5.3**

Table 4. Attack performance comparison on the MSCOCO dataset for image captioning task.

Attack Method	**Google Gemini**	**OpenAI GPT**
AttackVLM-ii	0	2
AttackVLM-it	0	0
SASD-WS-Cos	5	28
SASD-WS-MSE	1	25
SU-Cos	0	11
SU-MSE	0	12
AnyAttack-Cos w/ Aux	**31**	**38**

Table 5. Quantitative performance comparison on commercial VLMs. The reported values represent the ASR.

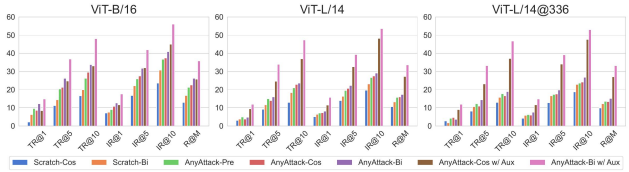

Figure 4. Ablation study for MSCOCO retrieval. The provided figure indicates that pre-training, auxiliary encoders, and the bidirectional objective each contribute materially to attack strength.

generation is amortized into a single forward pass. This is an important property for large-scale evaluation, where per-target optimization can dominate computational cost.

Cross-task transfer pattern. A consistent pattern across benchmarks is that the best fine-tuning variant depends on the downstream interface, while auxiliary encoders help in every setting. On retrieval, AnyAttack-Bi w/ Aux is the top model on all three CLIP victims, which is consistent with the fact that retrieval is evaluated by bidirectional ranking and therefore benefits from the symmetric contrastive objective used in adaptation. On SNLI-VE and captioning, the cosine-based variant remains competitive, but adding auxiliaries is still decisive: SNLI-VE accuracy rises from 17.5% to 44.8%, and CIDEr improves on all four captioning models. This behavior supports the interpretation that

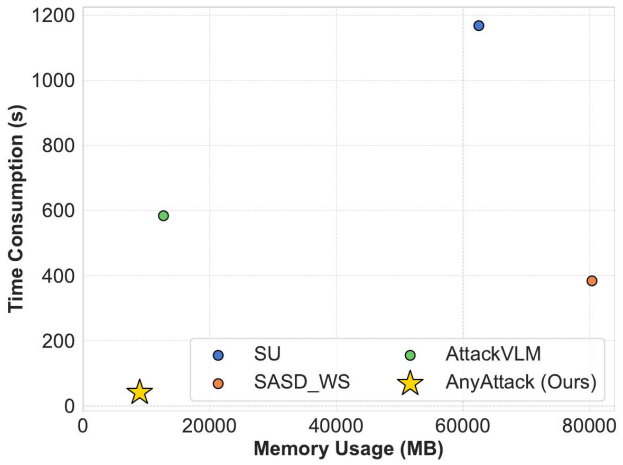

Figure 5. Runtime and memory comparison. Because AnyAttack-generates perturbations in one forward pass, it is markedly more efficient than iterative optimization-based attacks in the supplied comparison.

pre-training provides generic target-conditioned perturbation primitives, while task-matched fine-tuning determines how effectively those primitives transfer to each downstream VLM.

Effect of victim mismatch. Auxiliary supervision becomes more important as the evaluation model moves farther from the main surrogate configuration. In retrieval, adding auxiliaries to AnyAttack-Bi raises R@Mean from 26.23 to 35.88 on ViT-B/16, from 17.34 to 33.66 on ViT-L/14, and from 15.21 to 33.18 on ViT-L/14×336. The gain is therefore much larger on the larger CLIP variants than on the closest victim, which is consistent with the view that multi-encoder adaptation encourages perturbations aligned with features shared across backbones and resolutions. A similar pattern appears on captioning, where CIDEr increases from 7.0 to 20.3 on InstructBLIP and from 1.9 to 5.3 on MiniGPT-4. The commercial results remain below open-source retrieval performance, with 31% ASR on Gemini and 38% on GPT, showing both meaningful transfer and a persistent penalty from API-level mismatch.

Why carrier-target decoupling matters. The transfer pattern also supports the training design itself. Because pre-training and fine-tuning always pair a target embedding with shuffled carrier images, the decoder cannot rely on a fixed carrier structure or memorize narrow source-target correspondences. Instead, it must encode target semantics in a form that survives large appearance changes. This interpretation is consistent with the fact that gains persist from CLIP retrieval to SNLI-VE, captioning, and commercial APIs, even though these victims differ substantially from the pre-training setup. The smaller improvements on BLIP and the lower commercial ASR further indicate that this invariance is useful but not complete.

Discussion. The results suggest that targeted multimodal attacks benefit from the same scaling intuition that has powered representation learning: pre-train once on massive weak supervision, then adapt cheaply. At the same time, the experiments reveal limitations. The gains are largest on retrieval and somewhat smaller on some captioning models, especially BLIP, where the best baseline remains competitive on SPICE and close on CIDEr. This indicates that generative decoders may preserve some robustness depending on architecture and evaluation metric. Still, the consistent improvements across open-source and commercial models indicate that the learned perturbations capture transferable semantic directions rather than surrogate-specific artifacts. This points to shared semantic geometry as the main driver of transfer.

Boundary of transfer. The same results also clarify where the method remains constrained. Transfer is strongest when the downstream objective is close to the surrogate representation used in training, as seen in retrieval, and weaker under larger interface mismatch, as seen in the lower ASR on commercial APIs and the smaller gains on BLIP-style captioners. This suggests that AnyAttacklearns broadly reusable semantic attack directions, but not a fully model-agnostic one. For evaluation, this means a single benchmark is insufficient: measuring only retrieval would overestimate risk for some generative systems, while measuring only captioning would understate the vulnerability of contrastive VLM pipelines.

6. Conclusion

We presented AnyAttack, a targeted black-box attack that treats adversarial perturbation generation for VLMs as a foundation-modeling problem. By pre-training a perturbation decoder with a self-supervised contrastive objective on large-scale image data and then fine-tuning it with lightweight downstream losses, AnyAttackremoves the dependence on label supervision during large-scale training and reduces attack-time cost to a single forward pass. On the provided experimental suite, this design improves targeted transfer on retrieval, classification, captioning, and commercial VLMs, while also reducing time and memory relative to iterative baselines. The strongest gains appear on retrieval, but the positive transfer to SNLI-VE, captioning, and commercial APIs shows that the learned perturbation generator is not confined to one downstream interface.

Practical implication. These findings point to a practical security risk for modern VLM pipelines and suggest that robustness evaluations should account for large-scale learned perturbation generators, not only per-example optimization attacks.

References

[1] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International conference on machine learning*, pages 1597–1607. PMLR, 2020. 2, 3

[2] Wenliang Dai, Junnan Li, Dongxu Li, Anthony Tiong, Junqi Zhao, Weisheng Wang, Boyang Li, Pascale Fung, and Steven Hoi. InstructBLIP: Towards general-purpose vision-language models with instruction tuning. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. 1

[3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 4

[4] Yuxin Fang, Quan Sun, Xinggang Wang, Tiejun Huang, Xinlong Wang, and Yue Cao. Eva-02: A visual representation for neon genesis. *Image and Vision Computing*, 149:105171, 2024. 4

[5] Ian J Goodfellow, Jonathon Shlens, and Christian Szegedy. Explaining and harnessing adversarial examples. *arXiv preprint arXiv:1412.6572*, 2014. 1, 2

[6] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9729–9738, 2020. 2

[7] Chao Jia, Yinfei Yang, Ye Xia, Yi-Ting Chen, Zarana Parekh, Hieu Pham, Quoc Le, Yun-Hsuan Sung, Zhen Li, and Tom Duerig. Scaling up visual and vision-language representation learning with noisy text supervision. In *International conference on machine learning*, pages 4904–4916. PMLR, 2021. 2

[8] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*, pages 12888–12900. PMLR, 2022. 1

[9] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pages 19730–19742. PMLR, 2023. 1

[10] Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. Towards deep learning models resistant to adversarial attacks. *arXiv preprint arXiv:1706.06083*, 2017. 1, 2

[11] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021. 1, 2

[12] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, et al. Imagenet large scale visual recognition challenge. *International journal of computer vision*, 115 (3):211–252, 2015. 3

[13] Christoph Schuhmann, Richard Vencu, Romain Beaumont, Robert Kaczmarczyk, Clayton Mullis, Aarush Katta, Theo Coombes, Jenia Jitsev, and Aran Komatsuzaki. Laion-400m: Open dataset of clip-filtered 400 million image-text pairs. *arXiv preprint arXiv:2111.02114*, 2021. 2, 3

[14] Xiaosen Wang, Xuanran He, Jingdong Wang, and Kun He. Admix: Enhancing the transferability of adversarial attacks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 16158–16167, 2021. 2

[15] Xiaosen Wang, Zeliang Zhang, and Jianping Zhang. Structure invariant transformation for better adversarial transferability. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4607–4619, 2023. 1, 2, 4

[16] Zhipeng Wei, Jingjing Chen, Zuxuan Wu, and Yu-Gang Jiang. Enhancing the self-universality for transferable targeted attacks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12281–12290, 2023. 2, 4

[17] Cihang Xie, Zhishuai Zhang, Yuyin Zhou, Song Bai, Jianyu Wang, Zhou Ren, and Alan L Yuille. Improving transferability of adversarial examples with input diversity. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2730–2739, 2019. 1, 2

[18] Jiaming Zhang, Qi Yi, and Jitao Sang. Towards adversarial attack on vision-language pre-training models. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 5005–5013, 2022. 1, 2

[19] Qilong Zhang, Xiaodan Li, YueFeng Chen, Jingkuan Song, Lianli Gao, Yuan He, et al. Beyond imagenet attack: Towards crafting adversarial examples for black-box domains. In *International Conference on Learning Representations*, 2022. 1, 2

[20] Yunqing Zhao, Tianyu Pang, Chao Du, Xiao Yang, Chongxuan Li, Ngai-Man Man Cheung, and Min Lin. On evaluating adversarial robustness of large vision-language models. *Advances in Neural Information Processing Systems*, 36, 2024. 1, 2, 4

[21] Haonan Zheng, Wen Jiang, Xinyang Deng, and Wenrui Li. Sample-agnostic adversarial perturbation for vision-language pre-training models. In *Proceedings of the 32st ACM International Conference on Multimedia*, 2024. 2

[22] Ziqi Zhou, Shengshan Hu, Minghui Li, Hangtao Zhang, Yechao Zhang, and Hai Jin. Advclip: Downstream-agnostic adversarial examples in multimodal contrastive learning. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 6311–6320, 2023. 1, 2

[23] Deyao Zhu, Jun Chen, Xiaoqian Shen, Xiang Li, and Mohamed Elhoseiny. Minigpt-4: Enhancing vision-language understanding with advanced large language models. In *The Twelfth International Conference on Learning Representations*, 2024. 1